

Dhruva Aher

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SUMMARY

CS student at ASU building and shipping full-stack applications – REST APIs, React frontends, and cloud-deployed backends across distributed systems and AI. Built production systems including a platform processing 810K+ FEMA records and a deployed GitHub Marketplace app. AWS Cloud Practitioner and Azure AI certified. Seeking 2026 Summer SWE Internships

EDUCATION

Bachelor of Science in Computer Science

Expected May 2029

Arizona State University, Grand Challenges Scholars Program, Tempe, AZ

GPA: 4.0/4.0

- Relevant Coursework: Data Structures & Algorithms, Discrete Mathematics, Object-Oriented Programming, Principles of Programming
- Honors: Dean's List (Fall 2025)

TECHNICAL SKILLS & CERTIFICATIONS

Languages: Python, Java, JavaScript, TypeScript, SQL

Backend & APIs: FastAPI, Flask, Node.js, REST APIs, Async Programming, JWT Authentication, RBAC, SQLAlchemy, Alembic

Frontend: React, TypeScript, Zustand, TanStack Query, Vite

Databases & Systems: PostgreSQL, Redis, WebSockets, Celery

Data & Analytics: Pandas, ETL Pipelines, Data Modeling

Cloud & DevOps: Docker, Docker Compose, AWS (EC2, ECS, S3, RDS, CloudFront), Microsoft Azure, Terraform, GitHub Actions (CI/CD)

Concepts: Data Structures, Algorithms, OOP, Distributed Systems, Microservices, Agile

AI/ML: OpenAI API, Google Gemini API, Computer Vision

Certifications: AWS Cloud Practitioner, Microsoft Azure AI Essentials Professional

EXPERIENCE

ACM at ASU

March 2026 – Present

Officer 4 (Administration), Intern (Logistics)

- Promoted based on ownership and execution; leading development of ACM website and Discord, improving onboarding and communication for 250+ members.
- Supported GlobeHacks 2026 (ACM x GCN), managing Devpost workflows, resolving support tickets, and coordinating real-time communication for 250+ participants.
- Led Q&A support, workshops, and volunteer coordination, ensuring rapid issue resolution and smooth end-to-end event execution.

PROJECTS

Aura – Distributed Task Orchestration Platform

February 2026 – Present

- Built a distributed job queue system using **Node.js**, **Redis (Lua)**, and **PostgreSQL (Prisma)** with concurrent workers and retry workflows, processing 5K+ tasks.
- Designed fault-tolerant execution with atomic queue operations and idempotent state transitions to prevent duplicate jobs under concurrent load.
- Developed observability using **SSE** and **Redis pub/sub**, tracking 250–350 jobs/min throughput and p95 latency under load testing.
- Wrote **integration tests** covering queue edge cases and failure recovery, validating correctness under concurrency.

PRBeliefs – GitHub Marketplace App

April 2026 – Present

- Built an AI-powered pull request review agent using **Python**, **FastAPI**, and **LLaMA 3.3 via Groq API**, delivering code reviews within 1 second via GitHub Webhooks.
- Designed a rules engine enforcing coding standards with structured outputs, confidence scoring, and severity classification.
- Implemented authentication, webhook handling, and retry logic, ensuring reliable real-time execution with zero downtime.
- Developed **unit and integration tests** for webhook handling and authentication flows, ensuring production reliability.

FlowBoard – Full-Stack Developer

March 2026 – Present

- Architected a multi-tenant SaaS platform in **React**, **TypeScript**, and **FastAPI** supporting isolated workspaces, role-based access, and real-time document editing.
- Designed backend with **JWT authentication**, **RBAC**, and async **PostgreSQL** for scalable multi-user systems.
- Engineered real-time updates using **WebSockets** and **Redis**, optimizing client-server synchronization and state consistency.

Disaster Relief Response Gap Tracker

April 2026 – Present

- Built analytics dashboard processing 810K+ FEMA records across 3,000+ counties.
- Engineered ETL pipelines joining FEMA disbursement data with U.S. Census income records, revealing weak aid-to-income correlation ($r = 0.16$).
- Deployed infrastructure using **Docker**, **Terraform**, and **AWS (ECS, RDS, CloudFront)** with Redis caching.

LaundryLens – Assistive Computer Vision System

Aug 2025 – Dec 2025

- Built a Raspberry Pi-based system classifying 15+ clothing attributes with 90%+ accuracy using computer vision and OpenAI Vision API.

ACTIVITIES

ACM at ASU (Officer 4): Led technical workshops, hackathon operations (GlobeHacks 2026), and community initiatives, supporting 250+ participants and members.

Google Developer Student Club (GDSC): Participated in workshops on web development, cloud technologies, and AI/ML.

Hacker Devils: Competed in hackathons, building full-stack and AI-driven projects under time constraints.